CEN350 Database Management Systems Final Exam

14.06.2021

QUESTIONS

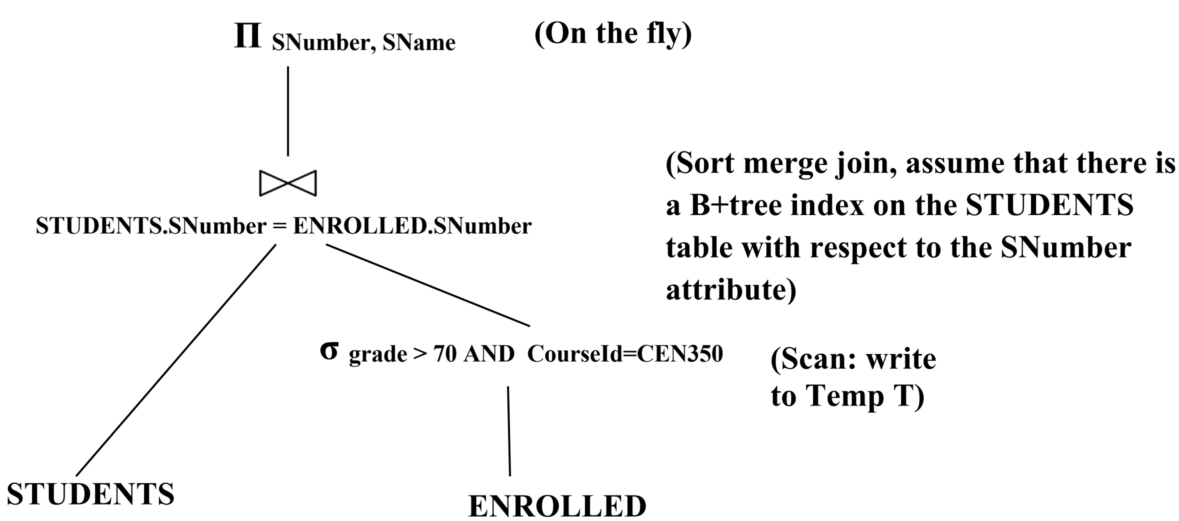
(***There are 4 questions in this exam, you should submit your document(s) until 14:30***)

1. **(30 points)** Consider the below given relation schema: write the following queries in **standard SQL**.

    STUDENTS(SNumber, SName, Department, Age)
    COURSES(CourseId, CourseName, Credit)
    ENROLLED(SNumber, CourseId, Semester, Grade)
    TEACHERS(TNumber, TName, CourseId)

    a) Find the SNumber, SName, and number of courses enrolled of the students who have enrolled at least 5 courses on “Spring_2021 semester”.
    b) Find the names of students who have grade 100 from the course named “Databases” for the “Spring_2021 semester”.
    c) Find the name and age of the oldest student enrolled in the course named “Databases” for the “Spring_2021 semester”.
    d) Find the names of teachers that do not offer any course for the “Spring_2021 semester”.

2. **(30 points)** You should use the relation schema given in the 1st question. For the below query plan, you are given that the STUDENTS table has 200 pages, the ENROLLED table has 10000 pages, and only the 20% of the tuples in the ENROLLED table satisfy the given selection condition. You can also assume that 11 buffer pages are available in memory.



    a) Estimate the cost of the above query plan in terms of number of disk accesses if generalized external merge sort (without tournament sort and without blocked disk access) is used for sorting.
    b) Write an SQL query which may be computed by the above query plan.

3. **(20 points)** Re-write the relation schema given in the 1st question assuming that **a teacher can give more than one course in one semester, and a course can be given by more than one teacher in one semester**. For example, more than one teacher can give CEN350 course for Spring 2021 semester. In

Spring 2021 semester, each teacher can give any number of courses. You should list the primary and foreign keys for each table and show that your relation schema does not have any redundant data. (You are not required write any SQL commands, you should just give the relation schema and list keys and functional dependencies for each relation)

4. **(20 points)** Consider the below schedule, where two transactions T1 and T2 run concurrently over the database objects A and B. In the below schedule R means Read operation, W means Write operation over the database object.

T1	**T2**
R(B)	
	W(B)
	W(A)
R(B)	
W(B)	
commit	
	commit

a) Explain any one problem in the above schedule. (It is enough to find any one problem in the above schedule).

b) Re-write the above schedule according to Strict 2 Phase Locking concurrency control protocol and explain whether the problem that you have found in the above question is solved or not.